Jihwan Oh

Stanford University, Stanford, CA, USA

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RESEARCH INTERESTS

Hardware-software co-design for efficient ML systems.

EDUCATION

– **Stanford University** Sep. 2026 –
Ph.D. Student in Electrical Engineering; Advisor: Prof. Thierry Tambe

– **Korea Advanced Institute of Science and Technology (KAIST)** Feb. 2019 – Feb. 2026
B.S. in School of Electrical Engineering (Primary Major), School of Computing (Double Major)
GPA: 4.07/4.3, *Summa Cum Laude*; Dean's List: Spring 2019, Spring 2024

– **Georgia Institute of Technology** Jan. 2025 – Jul. 2025
Exchange Program, School of Electrical and Computer Engineering; GPA: 4.0/4.0

PUBLICATIONS

Compute-Communication Overlap Is Not Free: A Cross-Layer Characterization in GPU LLM Workloads Sep. 2026
Jihwan Oh, Junkyum Kim, Seokjin Go, Jongse Park, Divya Mahajan
Best Paper Finalist; *Presented at IEEE International Symposium on Workload Characterization (IISWC) 2026*

Characterizing Compute-Communication Overlap in GPU-Accelerated Distributed Deep Learning: Performance and Power Implications May. 2025
Seonho Lee, ***Jihwan Oh***, Junkyum Kim, Seokjin Go, Jongse Park, Divya Mahajan
Poster presented at ISPASS 2025, arXiv:2507.03114

RESEARCH EXPERIENCE

Tambe Lab, Stanford University Sep. 2026 – Present
Ph.D. Researcher (Advisor: Prof. Thierry Tambe)
– Investigating quantization strategies and their performance implications for efficient ML workloads.

Systems Infrastructure and Architecture Research Lab, Georgia Tech Jan. 2025 – Aug. 2026
Researcher (Advisor: Prof. Divya Mahajan)
– Used GPU profiling to characterize compute-communication overlap in distributed LLM training, identifying hardware-level overheads that can make aggressive overlap degrade performance and power efficiency.
– Designed a TMA-driven NCCL communication protocol that stages data in otherwise unused shared memory, achieving comparable or higher bandwidth with roughly 50% fewer SMs in microbenchmarks.

Computer Architecture and Systems Lab, KAIST Sep. 2024 – Dec. 2024
Undergraduate Researcher (Advisor: Prof. Jongse Park)
– Analyzed NPU-PIM load imbalance during LLM inference on *NeuPIMs* and designed FFN workload redistribution to improve utilization and end-to-end performance.
– Surveyed quantization methods for PIM architectures and benchmarked model accuracy across multiple models.

ADDITIONAL EXPERIENCE

Teaching Assistant, KAIST
– Introduction to Programming (Fall 2025); Immersion Camp (Oct. 2023 – Feb. 2024).

Software Developer, Republic of Korea Air Force Aug. 2021 – May. 2023
– Developed a VR flight-training simulator in Unreal Engine 4 and C++; selected to present it at the 2022 ROKAF Information & Communication Development Conference.

AWARDS & SCHOLARSHIPS

- **Stanford Thrive Fellow**, Stanford University 2026
- **Best Paper Finalist**, IEEE IISWC 2026 2026
- **Next-Generation Engineer Award: Highest Distinction**, IPESK 2025
  Awarded the top distinction as the single highest-ranked recipient in the department.
- **uArch Mentoring Workshop Full Travel Grant**, uArch @ ISCA 2025 2025
- **National Science and Technology Scholarship**, Korea Government 2021 – 2024

TECHNICAL SKILLS

- **Programming:** C/C++, Python, Rust, SystemVerilog
- **ML/GPU Systems:** CUDA, PyTorch, NCCL, CUTLASS, cuBLAS, Megatron-LM, Megatron-DeepSpeed, Nsight Compute, Nsight Systems